

Today's Agenda:-

Starting 7:05

1. Search in a row-wise
k col. wise sorted matrix.
2. Sum of all submatrices.
3. Find first missing Integer.

Revision:-

1. Max Subarray Sum $\rightarrow$ Kadane.

2. Prefix Sum.

3. $O(Q + N)$.

Problem 1: Given a row wise & column wise sorted matrix, search for a given element K .

4×4

-5	-2	1	13
-4	0	3	14
-3	2	5	18
2	6	10	20

$N \times M$

ex:- 1) $K = 10$ true.

2) $K = 15$ false.

Brute Force:-

-5	-2	1	13
-4	0	3	14
-3	2	5	18
2	6	10	20

$N \times M$

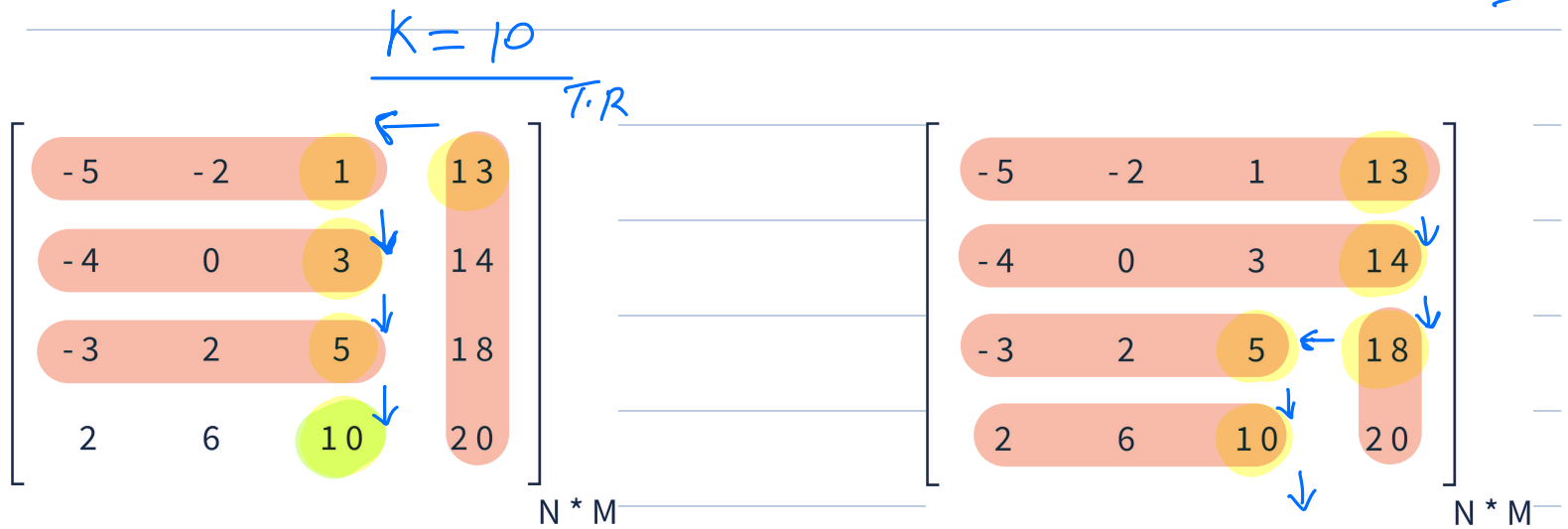
Iterate on every cell of the matrix & check if K is present or not.

T.C $\rightarrow O(N \times m)$

Q:- 1

Optimisation

In every iteration, reject either a row or a column.



return true.

$K = 15$.

return false.

Approach:-

1. Start at the top right cell
bottom left
2. If $A[i][j] > K$ (reject col or $\leftarrow$)
else if $A[i][j] < K$ (reject row or $\downarrow$)
3. Repeat until K is found or you have gone out of bound.

$N \times m$

Code

$i = 0; j = m - 1;$

while ($i < N$ & $j \geq 0$) {

if ($A[i][j] == k$) {
return true;

}

else if ($A[i][j] > k$) {

$j--;$

else {

$i++;$

}

}

return false;

T.C $\rightarrow$

$O(N + m)$

S.C $\rightarrow O(1)$

-5	-2	1	13
-4	0	3	14
-3	2	5	18
2	6	10	20

N * M

Sub - Matrices

$$N^2 m^2$$

$$O(N^2)$$

- Contiguous part of a matrix

Total Submatrices

	0	1	2	3
0	1	2	3	4
1	5	6	7	8
2	9	10	11	12
3	13	14	15	16

$$N(N+1) \times m(m+1)$$

$$4$$

$$3 \times 1$$

How to uniquely identify a submatrix/rectangle?

TL

TR

BL

BR

Using:-

$$(1) TL \times BR$$

OR

$$(2) TR \times BL$$

$$N^2 m^2$$

Problem 2 :- Given a matrix of size $N \times m$. Cal. k point the final sum of the sum of all its submatrices.

$$\underline{\underline{2 \times 3}}$$

ex:-

	0	1	2
0	4	9	6
1	5	-1	2

$1 \times 1 \checkmark$	$2 \times 1 \checkmark$
$1 \times 2 \checkmark$	$2 \times 2 \checkmark$
$1 \times 3 \checkmark$	$2 \times 3 \checkmark$

$$\text{ans} = \underline{\underline{164}}$$

Submatrices

1 * 1

$$\begin{bmatrix} 4 \end{bmatrix} \quad 4$$

1 * 2

$$\begin{bmatrix} 4 & 9 \end{bmatrix} \quad 13$$

1 * 3

$$\begin{bmatrix} 4 & 9 & 6 \end{bmatrix} \quad 19$$

$$\begin{bmatrix} 9 \end{bmatrix} \quad 9$$

$$\begin{bmatrix} 9 & 6 \end{bmatrix} \quad 15$$

$$\begin{bmatrix} 5 & -1 & 2 \end{bmatrix} \quad 6$$

$$\begin{bmatrix} 6 \end{bmatrix} \quad 6$$

$$\begin{bmatrix} 5 & -1 \end{bmatrix} \quad 4$$

$$\begin{bmatrix} 5 \end{bmatrix} \quad 5$$

$$\begin{bmatrix} -1 & 2 \end{bmatrix} \quad 1$$

$$\begin{bmatrix} -1 \end{bmatrix} \quad -1$$

$$\begin{bmatrix} 2 \end{bmatrix} \quad 2$$

2 * 1

$$\begin{bmatrix} 4 \\ 5 \end{bmatrix} \quad 9$$

2 * 2

$$\begin{bmatrix} 4 & 9 \\ 5 & -1 \end{bmatrix} \quad 17$$

2 * 3

$$\begin{bmatrix} 4 & 9 & 6 \\ 5 & -1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 9 \\ -1 \end{bmatrix} \quad 8$$

$$\begin{bmatrix} 9 & 6 \\ -1 & 2 \end{bmatrix} \quad 16$$

25

$$\begin{bmatrix} 6 \\ 2 \end{bmatrix} \quad 8$$

Brute Force

Consider every submatrix

$$N^2 m^2 \quad Nm = \underline{O(N^3 m^3)}$$

↓ Applying psom

$$O(N^2 m^2).$$

Optimisation

Identify

the total no.

of submatrices on element at
idx (i, j) contributing.

ex:-

(2, 2)

	0	1	2	3
0	✓	✓	✓	
1	✓	✓	✓	
2	✓	✓	✓	✓
3			✓	✓
4			✓	✓

Total Submatrices
m which $(2, 2)$
contributing.

$$= 9 \times 6$$

$$= \underline{\underline{54}};$$

Quiz 3

$$N=4 ; m=5$$

$$(1,2)$$

	0	1	2	3	4
0	✓	✓	✓		
1	✓	✓	✓	✓	✓
2			✓	✓	✓
3			✓	✓	✓

$$T.L = (i+1) \times (j+1)$$

$$= (2) \times (3)$$

$$B.R = (N-i) \times (m-j)$$

$$= (4-1) \times (5-2)$$

$$= 3 \times 3 = 9$$

Generalisation

For any (i, j)

$[0, j]$

	0	...	j	...	m-1
0	✓	✓	✓	✓	
⋮	✓	✓	✓	✓	
⋮	✓	✓	✓	✓	
i	✓	✓	✓	✓	✓
⋮				✓	✓
N-1				✓	✓

Total possibilities of

① T.L.

$$= (j+1) \times (i+1)$$

② B.R

$$= (m-j) \times (N-i)$$

$$[j, m-1] = (m-j+1)$$

$$[i, N-1] = (N-1-i+1)$$

$$\begin{aligned} \therefore \text{Total Submatrices in which} \\ A[i][j] \text{ is contributing} \\ &= T \cdot L \times B \cdot R \\ &= (j+1) \times (i+1) \\ &\quad \times (m-j) \times (n-i) \end{aligned}$$

$$\therefore \text{Total contribution of } A[i][j]$$

$$= A[i][j] \times (j+1) \times (i+1) \times (m-j) \times (n-i)$$

Code

```
int ans = 0;
```

```
for (i = 0; i < N; i++) {  
    for (j = 0; j < m; j++) {  
        ans += A[i][j] * (j+1) * (i+1)  
              * (m-j) * (N-i)  
    }  
}  
return ans;
```

$T.C \rightarrow O(N \times m)$

$S.C \rightarrow O(1)$

Break till 8:38

1, 2, 3, 4, ...

< Question > : Find the first missing natural number.

N = length of array

Example 1: arr[5]: [3 -1 1 2 7] ans = 4.

Example 2: arr[7]: [9 2 6 4 -8 1 3] ans = 5

Example 3: arr[6]: [1 0 -5 -6 4 2] ans = 3.

Example 4: arr[6]: [1 2 5 6 4 3] ans = 7.

Example 5: arr[4]: [1 2 3 4] ans = 5.

70.

Q: 4 { 5, 3, 1, -1, -2, -4, 7, 2 } ans = 4.

ans $\longrightarrow$ [1, N+1]

Idea 1:- For all values $[1, N+1]$, iterate on the array to find out whether it exists or not.

$$\underline{T.C \rightarrow O(N^2)}$$

$$S.C \rightarrow \underline{O(1)}.$$

Idea 2:- Sort the entire array.

$\{ 5, 3, 1, -1, -2, -4, 7, 2 \}$

↓ sort

$\{ -4, -2, -1, 1, 2, 3, 5, 7 \}$

↓ $ans = 4.$

$$T.C \rightarrow O(N \log N) ; S.C \rightarrow \underline{O(N)}.$$

Idea 3:- Try to place the available N.N. at their correct position.

{ 1 2 3 4 5 6 7 8 }

0 1 2 3 4 5 6 7

i^{th} N.N. $\longrightarrow$ $i-1^{\text{th}}$ index.

ex:- { ~~-8~~, ~~-7~~, ~~1~~, 10, ~~3~~, 8, ~~2~~ }

0 1 2 3 4 5 6

1 2 3

i

$[1, N]$

≤ 1 OK

$> N$

ans = 4.

{ ~~5~~, ~~3~~, ~~1~~, -1, ~~-2~~, -4, 7, ~~2~~, 3 }

0 1 2 3 4 5 6 7

~~-2~~ ~~1~~ 3 5 ~~-2~~

1 2

ans = 4.

Approach:-

1. In 1st iteration, ^{try} to bring any valid ^{N-N} ^{natural number} in range $[1, N]$ at its correct position
 $i^{\text{th}} \rightarrow i-1$ ^{index} ^{index}

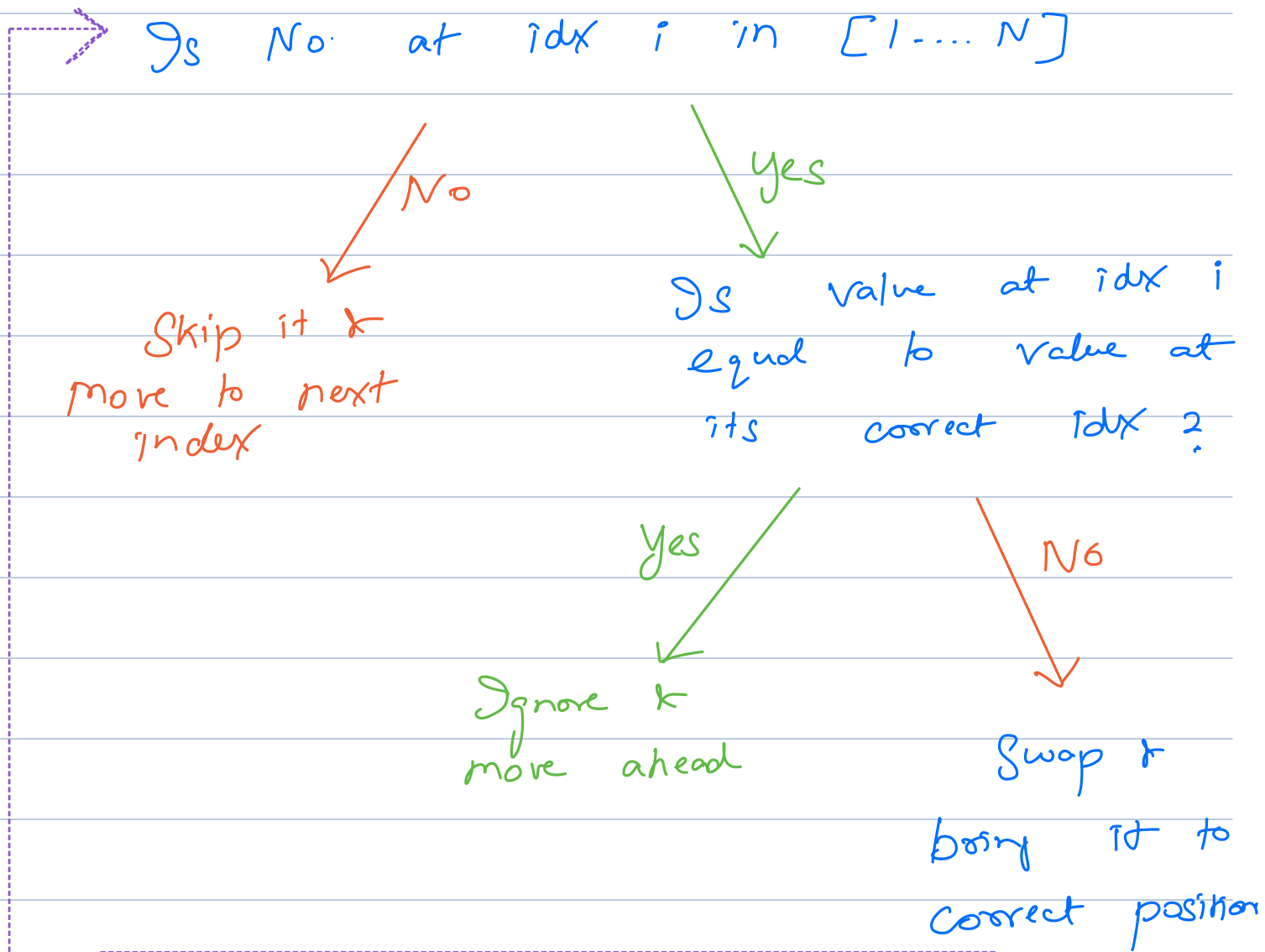
2. Iterate the ^{array} to check
if $arr[i] == i+1$
If not, $i+1$ is the answer.

After iterating, if no violation found; return $N+1$.

i
↓
 $idx = 6$
 $val = 4$

↗ updated only
 $A[A[i] - 1]$

flow diagram



$f = 6$
 $val = 4.$

Code

step-1

int i = 0;

while (i < N) {

if (A[i] > 1 && A[i] ≤ N
&&

A[i] != A[A[i]-1]) {

swap (A[i], A[A[i]-1];

else {

i++;

}

step-2

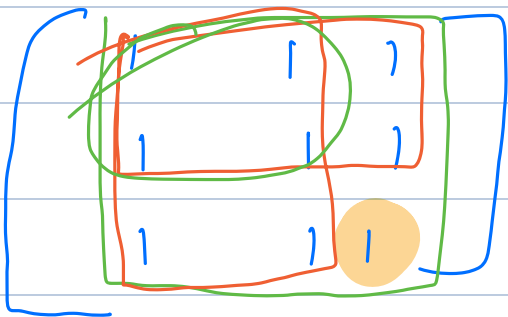
for (i = 0; i < N; i++) {

if (A[i] != i+1) {

return i+1;

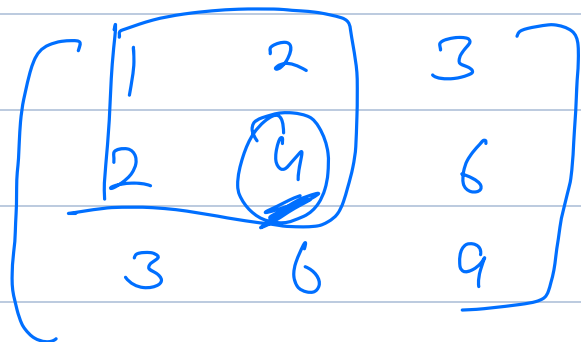
set $N+1$;

$$\underline{\underline{T.C \rightarrow O(N)}}$$



$$= \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}$$

$$9 - 6 - 6 + 4 = 1$$



$$- (\hat{i}, \hat{j}) ; \quad (\hat{i}-1, \hat{j}-1)$$

$$i \xrightarrow{[1,5]}$$

$$\{ \underset{0}{6}, \underset{1}{13}, \underset{2}{289}, \underset{3}{12}, \underset{4}{22} \}$$